

Credit Risk Classification Using Machine Learning on the German Credit Dataset

1 Objective:

- To develop and evaluate multiple supervised machine learning models for predicting credit risk (good vs. bad credit) using the German Credit Dataset.
 - The goal is to identify the most effective model using performance metrics such as ROC AUC, accuracy, precision, recall, and F1-score, both via an 80-20 train-test split and 5-fold cross-validation.
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2 Methodology:

- Data Source: Used the publicly available German Credit Dataset containing 1000 loan applicants described by 20 categorical and numerical features, labeled as good or bad credit risk.
- Exploratory Data Analysis (EDA):
 - Inspected feature distributions using histograms and boxplots.
 - Evaluated class balance (700 good, 300 bad).
 - Checked for missing values and correlations among numerical features using a heatmap.
- Data Preprocessing:
 - Handled categorical variables using one hot encoding.
 - Scaled numerical features where necessary to improve model performance.
- Model Evaluation Approach:
 - 80-20 Train-Test Split:
 - * Stratified split with 80% training and 20% testing.
 - * Pipeline with preprocessing and classification for each model.
 - * Evaluated using:
 - ROC AUC
 - Accuracy, Precision, Recall, and F1-score
 - ROC curve visualization.

- 5-Fold Cross-Validation:
 - Used StratifiedKFold to ensure balanced classes across folds.
 - Evaluated mean AUC across folds for each model.
 - Models Implemented:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Random Forest Classifier
 - XGBoost Classifier
 - K-Nearest Neighbors (KNN)
 - Tools and Libraries: Python, with scikit-learn, xgboost, pandas, numpy, matplotlib, and seaborn.
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3 Key Findings:

- From 80-20 Train-Test Split:
 - Logistic Regression delivered the highest AUC (~ 0.819) and balanced overall performance, confirming its effectiveness for linear classification in this context.
 - Random Forest and XGBoost had comparable AUCs (~ 0.811), but XGBoost showed better recall (0.53), making it more suitable for detecting risky applicants.
 - SVM maintained high precision but had low recall (0.28), limiting its utility in identifying bad credit cases.
 - KNN underperformed (AUC ~ 0.689), consistent with challenges in high dimensional tabular data.
 - From 5-Fold Cross-Validation:
 - Logistic Regression had the best average AUC (~ 0.787), again confirming its stability across folds.
 - SVM and Random Forest followed closely (AUC ~ 0.781), showing consistent but slightly lower performance.
 - XGBoost slightly underperformed (AUC ~ 0.773), indicating higher variance and sensitivity to data splits.
 - KNN remained the least effective model (AUC ~ 0.684), again confirming its unsuitability for this task.
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4 Conclusions:

- The project effectively demonstrates both holdout based and cross validated model evaluation for credit risk prediction.
- Logistic Regression emerged as the most consistent and interpretable model.
- XGBoost and Random Forest offer powerful alternatives with good generalization but may require more careful tuning.
- SVM had strong precision but lower recall, indicating trade offs in minority class detection.
- KNN is not well-suited to this type of data.

5 Code and analysis:

[]:

Libraries

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split, StratifiedKFold, \
    ↪ cross_val_score
from sklearn.metrics import accuracy_score, precision_score, recall_score, \
    ↪ f1_score, roc_auc_score, roc_curve, classification_report
```

Dataset

German Credit Dataset Summary:

- Source: Prof. Dr. Hans Hofmann, University of Hamburg. Available at <https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>.
- Instances: 1000
- Attributes: 20 features (7 numerical, 13 categorical). The attributes are: Status, Duration, CreditHistory, Purpose, CreditAmount, Savings, EmploymentSince, InstallmentRate, PersonalStatusSex, OtherDebtors, ResidenceSince, Property, Age, OtherInstallmentPlans, Housing, ExistingCredits, Job, LiablePeople, Telephone, ForeignWorker.
- Label (Target):
 - 1: Good credit
 - 2: Bad credit
- Description: This dataset classifies people described by a set of attributes as good (1) or bad (2) credit risks. The original version (german.data) includes categorical variables with symbolic codes. The numeric version (german.data-numeric) is preprocessed for use with algorithms requiring numeric input.

```
[2]: columns = [
    'Status', 'Duration', 'CreditHistory', 'Purpose', 'CreditAmount', 'Savings',
    'EmploymentSince', 'InstallmentRate', 'PersonalStatusSex', 'OtherDebtors',
    'ResidenceSince', 'Property', 'Age', 'OtherInstallmentPlans', 'Housing',
```

```

    'ExistingCredits', 'Job', 'LiablePeople', 'Telephone', 'ForeignWorker',
    ↪ 'Target'
]

df = pd.read_csv('german.data', sep=' ', header=None, names=columns)
df.head()

```

```

[2]:  Status  Duration  CreditHistory  Purpose  CreditAmount  Savings  \
0     A11         6             A34     A43         1169     A65
1     A12        48             A32     A43         5951     A61
2     A14        12             A34     A46         2096     A61
3     A11        42             A32     A42         7882     A61
4     A11        24             A33     A40         4870     A61

    EmploymentSince  InstallmentRate  PersonalStatusSex  OtherDebtors  ...  \
0                A75                4                A93         A101  ...
1                A73                2                A92         A101  ...
2                A74                2                A93         A101  ...
3                A74                2                A93         A103  ...
4                A73                3                A93         A101  ...

    Property Age  OtherInstallmentPlans  Housing  ExistingCredits  Job  \
0     A121  67             A143     A152         2     A173
1     A121  22             A143     A152         1     A173
2     A121  49             A143     A152         1     A172
3     A122  45             A143     A153         1     A173
4     A124  53             A143     A153         2     A173

    LiablePeople  Telephone  ForeignWorker  Target
0              1      A192             A201      1
1              1      A191             A201      2
2              2      A191             A201      1
3              2      A191             A201      1
4              2      A191             A201      2

[5 rows x 21 columns]

```

```

[3]: # Convert target to binary: 1 = good, 2 = bad → make 1 = 0, 2 = 1
df['Target'] = df['Target'].map({1: 0, 2: 1})
# Separate features and target
X = df.drop('Target', axis=1)
y = df['Target']

# Encode categorical features
categorical_cols = X.select_dtypes(include='object').columns.tolist()
numerical_cols = X.select_dtypes(exclude='object').columns.tolist()

```

Check for null values

```
[4]: df.isnull().sum()
```

```
[4]: Status          0
     Duration        0
     CreditHistory   0
     Purpose         0
     CreditAmount    0
     Savings         0
     EmploymentSince 0
     InstallmentRate 0
     PersonalStatusSex 0
     OtherDebtors    0
     ResidenceSince  0
     Property        0
     Age            0
     OtherInstallmentPlans 0
     Housing         0
     ExistingCredits 0
     Job            0
     LiablePeople    0
     Telephone       0
     ForeignWorker   0
     Target          0
     dtype: int64
```

Basic info of the dataset

```
[5]: print("Dataset Shape:")
     print(df.shape)
     print("\n Data Types:")
     print(df.dtypes)
     print("\n Descriptive Statistics:")
     print(df.describe())
```

Dataset Shape:
(1000, 21)

Data Types:

Status	object
Duration	int64
CreditHistory	object
Purpose	object
CreditAmount	int64
Savings	object
EmploymentSince	object
InstallmentRate	int64
PersonalStatusSex	object

```

OtherDebtors          object
ResidenceSince        int64
Property              object
Age                   int64
OtherInstallmentPlans object
Housing               object
ExistingCredits        int64
Job                   object
LiablePeople           int64
Telephone              object
ForeignWorker          object
Target                int64
dtype: object

```

Descriptive Statistics:

	Duration	CreditAmount	InstallmentRate	ResidenceSince \
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	20.903000	3271.258000	2.973000	2.845000
std	12.058814	2822.736876	1.118715	1.103718
min	4.000000	250.000000	1.000000	1.000000
25%	12.000000	1365.500000	2.000000	2.000000
50%	18.000000	2319.500000	3.000000	3.000000
75%	24.000000	3972.250000	4.000000	4.000000
max	72.000000	18424.000000	4.000000	4.000000

	Age	ExistingCredits	LiablePeople	Target
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	35.546000	1.407000	1.155000	0.300000
std	11.375469	0.577654	0.362086	0.458487
min	19.000000	1.000000	1.000000	0.000000
25%	27.000000	1.000000	1.000000	0.000000
50%	33.000000	1.000000	1.000000	0.000000
75%	42.000000	2.000000	1.000000	1.000000
max	75.000000	4.000000	2.000000	1.000000

Class Distribution

```

[6]: print(df['Target'].value_counts())
      df['Target'].value_counts().plot(kind='bar')

```

```

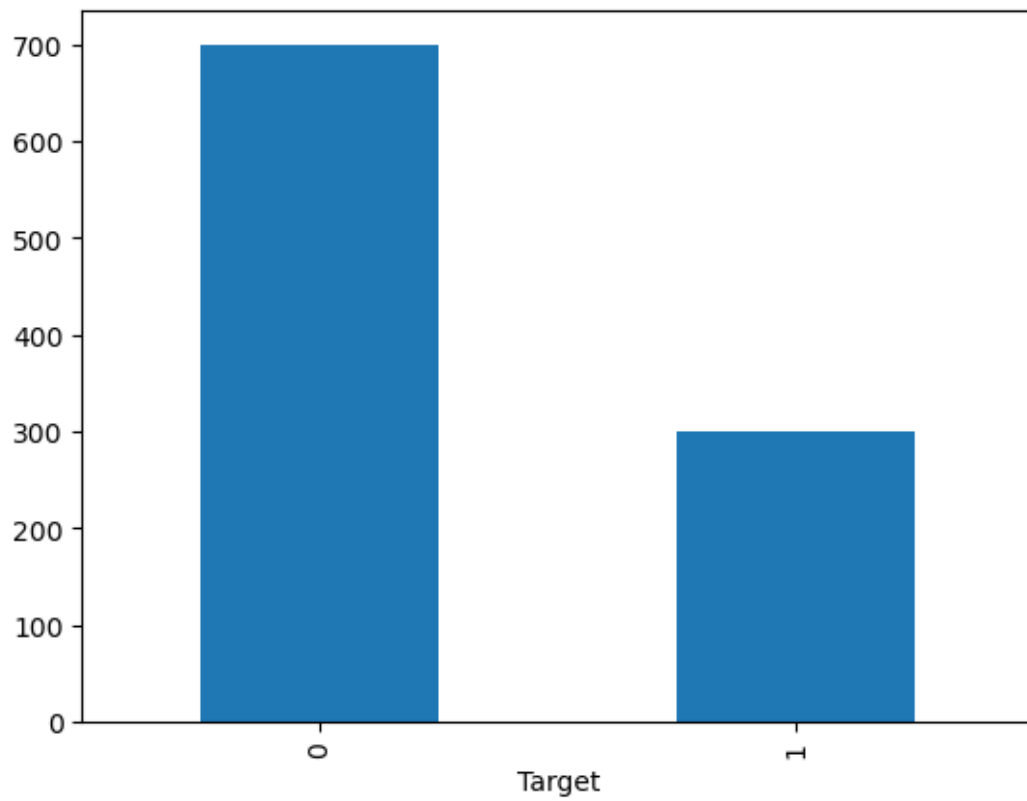
Target
0      700
1      300
Name: count, dtype: int64

```

```

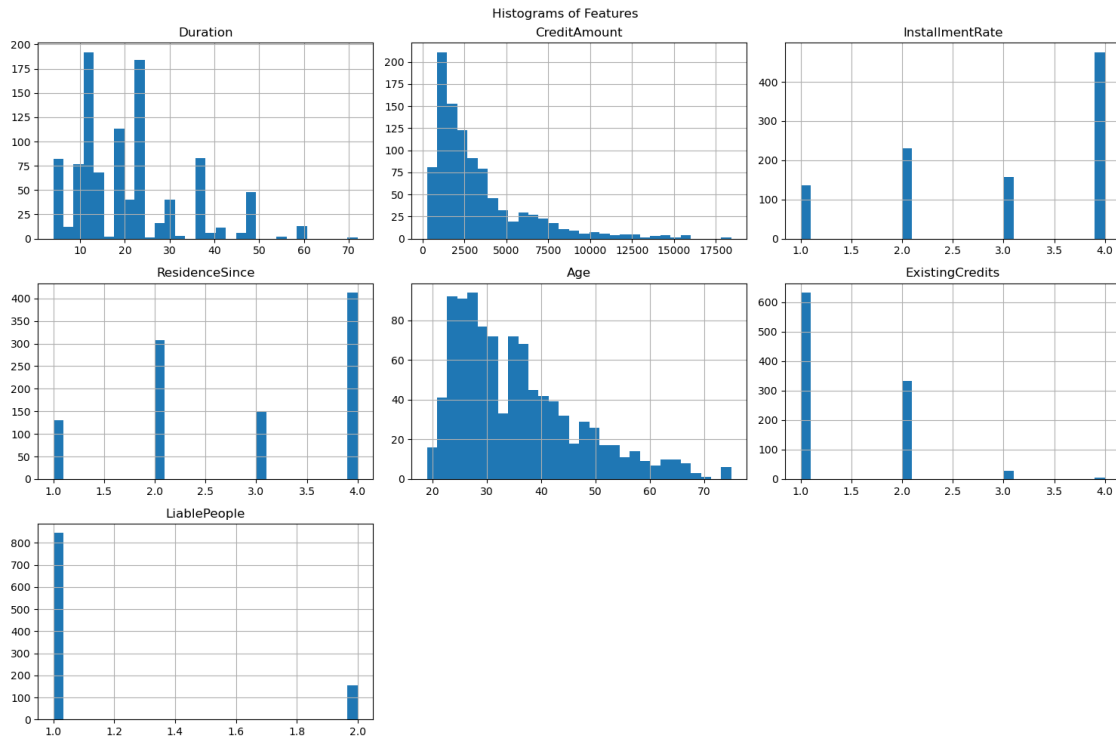
[6]: <Axes: xlabel='Target'>

```



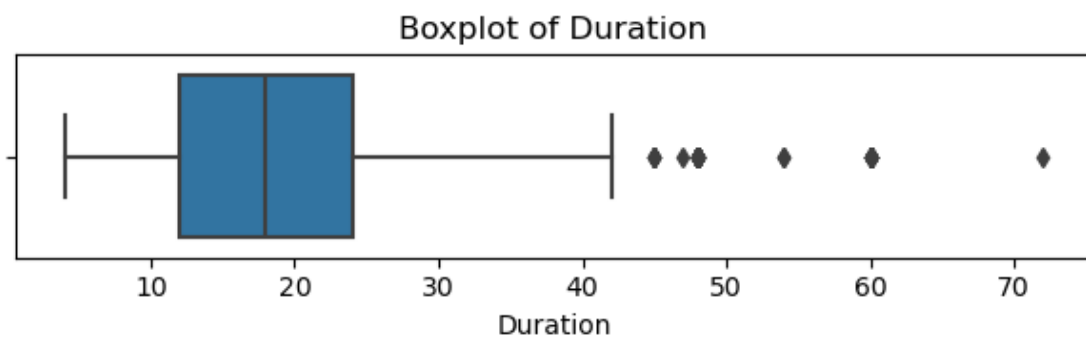
Feature distributions

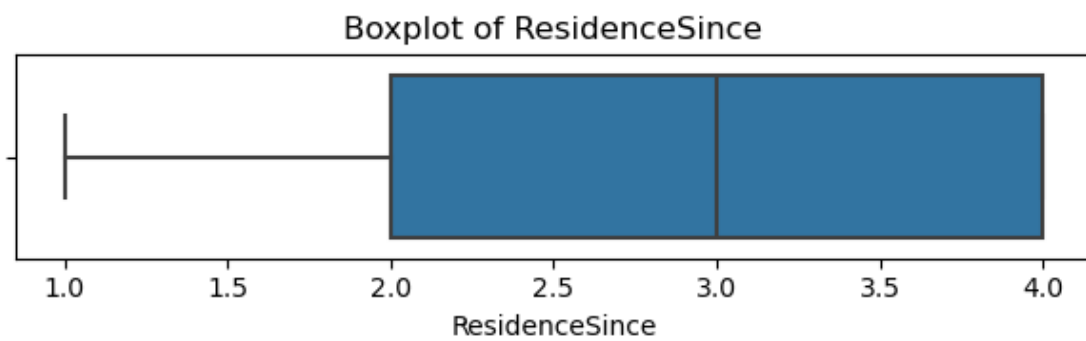
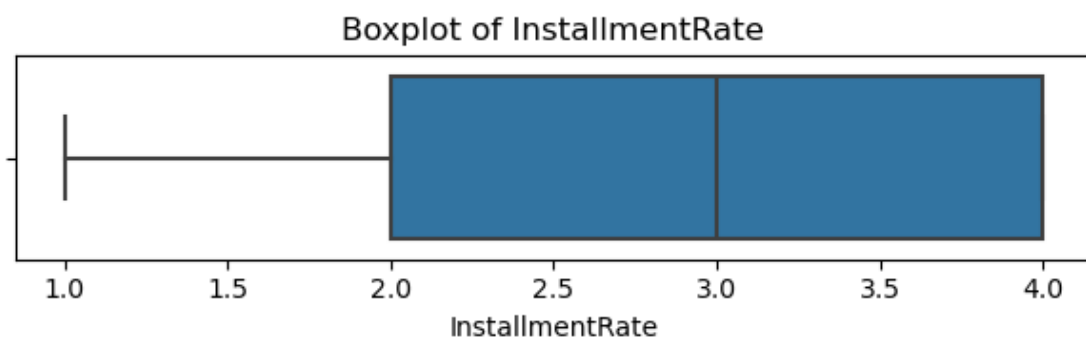
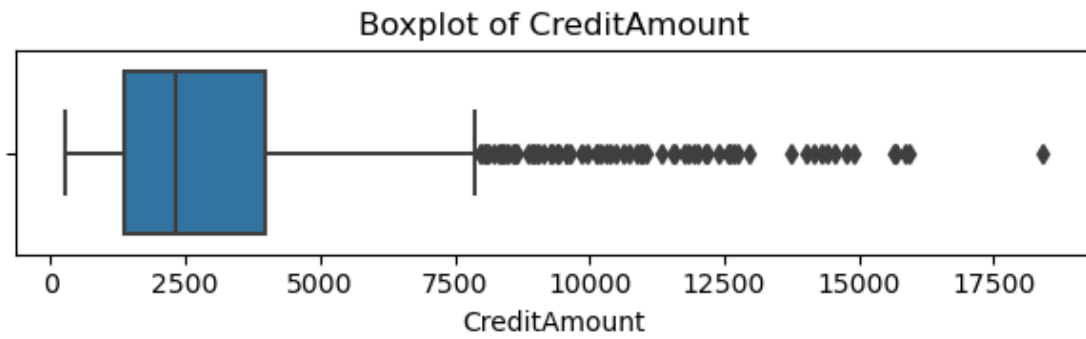
```
[7]: df[numerical_cols].hist(bins=30, figsize=(15, 10))  
plt.suptitle("Histograms of Features")  
plt.tight_layout()  
plt.show()
```

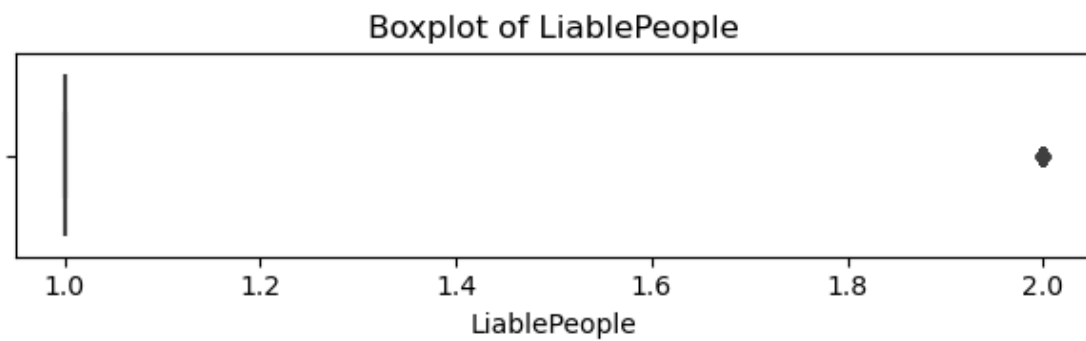
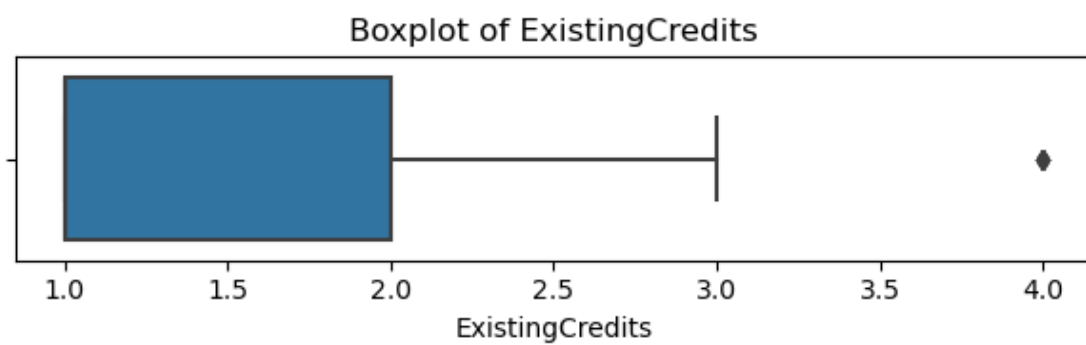
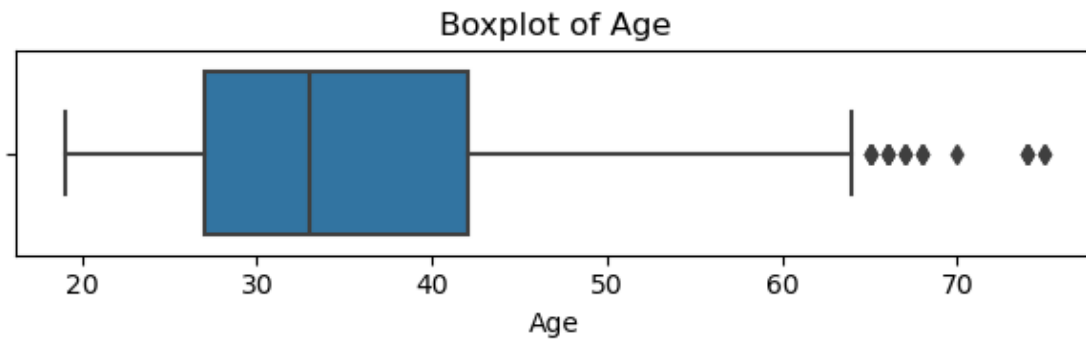


Boxplots for outlier detection

```
[8]: for col in numerical_cols:
    plt.figure(figsize=(6, 2))
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.tight_layout()
    plt.show()
```

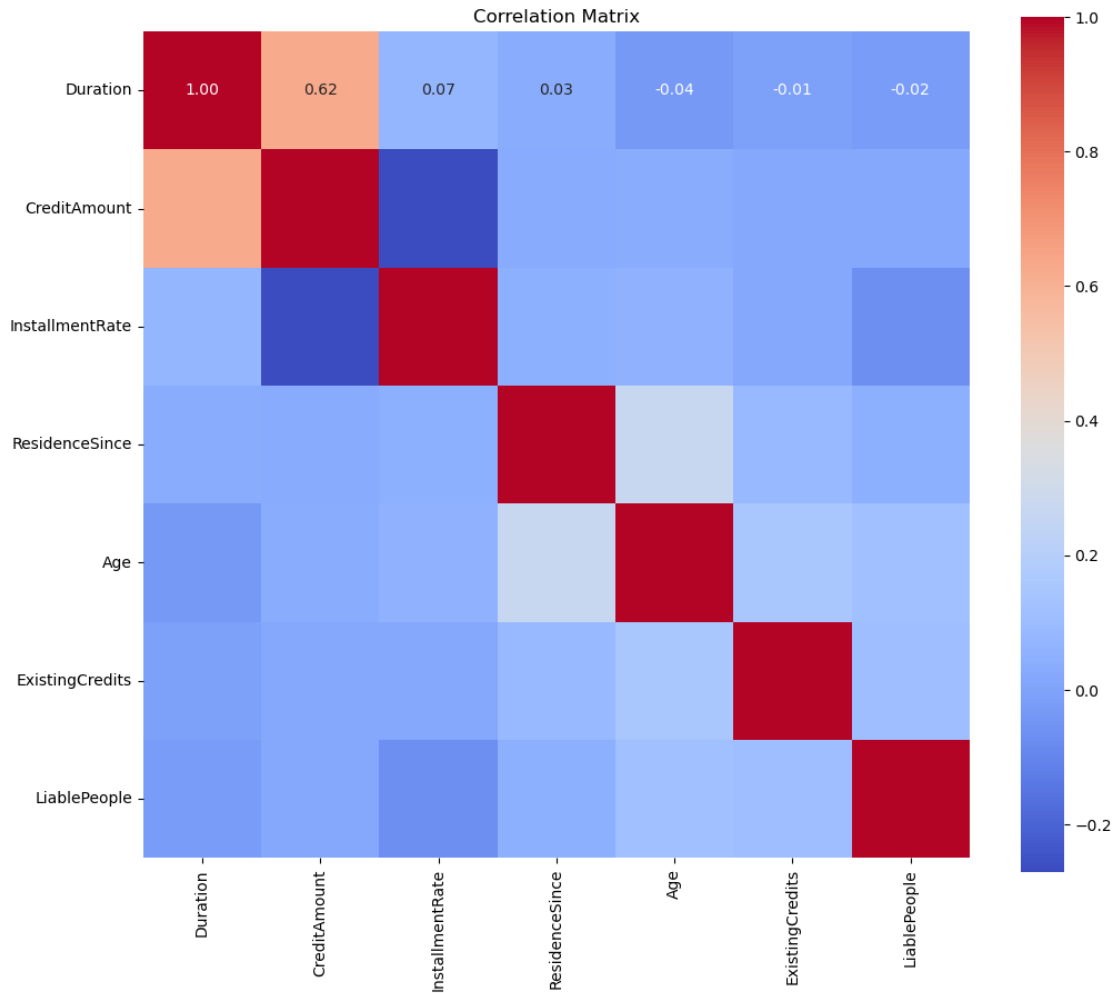






Correlation heatmap

```
[9]: plt.figure(figsize=(12, 10))
corr_matrix = df[numerical_cols].corr()
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm", square=True)
plt.title("Correlation Matrix")
plt.show()
```



Create preprocessing pipeline

```
[10]: preprocessor = ColumnTransformer([
    ('cat', OneHotEncoder(drop='first'), categorical_cols),
    ('num', StandardScaler(), numerical_cols)
])
```

Define models

```
[11]: models = {
    'Logistic Regression': LogisticRegression(max_iter=1000),
    'SVM': SVC(probability=True),
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=100),
    'XGBoost': XGBClassifier(eval_metric='logloss', random_state=100),
    'KNN': KNeighborsClassifier(n_neighbors=5)
}
```

```

[12]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↳stratify=y, random_state=100)

plt.figure(figsize=(10, 7))
results = {}

for name, model in models.items():
    print(f"\n {name} : ")

    # Pipeline with preprocessing and model
    pipeline = Pipeline(steps=[
        ('preprocessor', preprocessor),
        ('classifier', model)
    ])

    pipeline.fit(X_train, y_train)

    y_pred = pipeline.predict(X_test)
    y_proba = pipeline.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    auc = roc_auc_score(y_test, y_proba)

    results[name] = {
        'Accuracy': acc,
        'Precision': prec,
        'Recall': rec,
        'F1 Score': f1,
        'AUC': auc
    }

    print(classification_report(y_test, y_pred, digits=4))

    fpr, tpr, _ = roc_curve(y_test, y_proba)
    plt.plot(fpr, tpr, label=f'{name} (AUC = {auc:.2f})')

plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve Comparison (80-20 Split)')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```

Logistic Regression :

	precision	recall	f1-score	support
0	0.8052	0.8857	0.8435	140
1	0.6522	0.5000	0.5660	60
accuracy			0.7700	200
macro avg	0.7287	0.6929	0.7048	200
weighted avg	0.7593	0.7700	0.7603	200

SVM :

	precision	recall	f1-score	support
0	0.7584	0.9643	0.8491	140
1	0.7727	0.2833	0.4146	60
accuracy			0.7600	200
macro avg	0.7656	0.6238	0.6318	200
weighted avg	0.7627	0.7600	0.7187	200

Random Forest :

	precision	recall	f1-score	support
0	0.7765	0.9429	0.8516	140
1	0.7333	0.3667	0.4889	60
accuracy			0.7700	200
macro avg	0.7549	0.6548	0.6703	200
weighted avg	0.7635	0.7700	0.7428	200

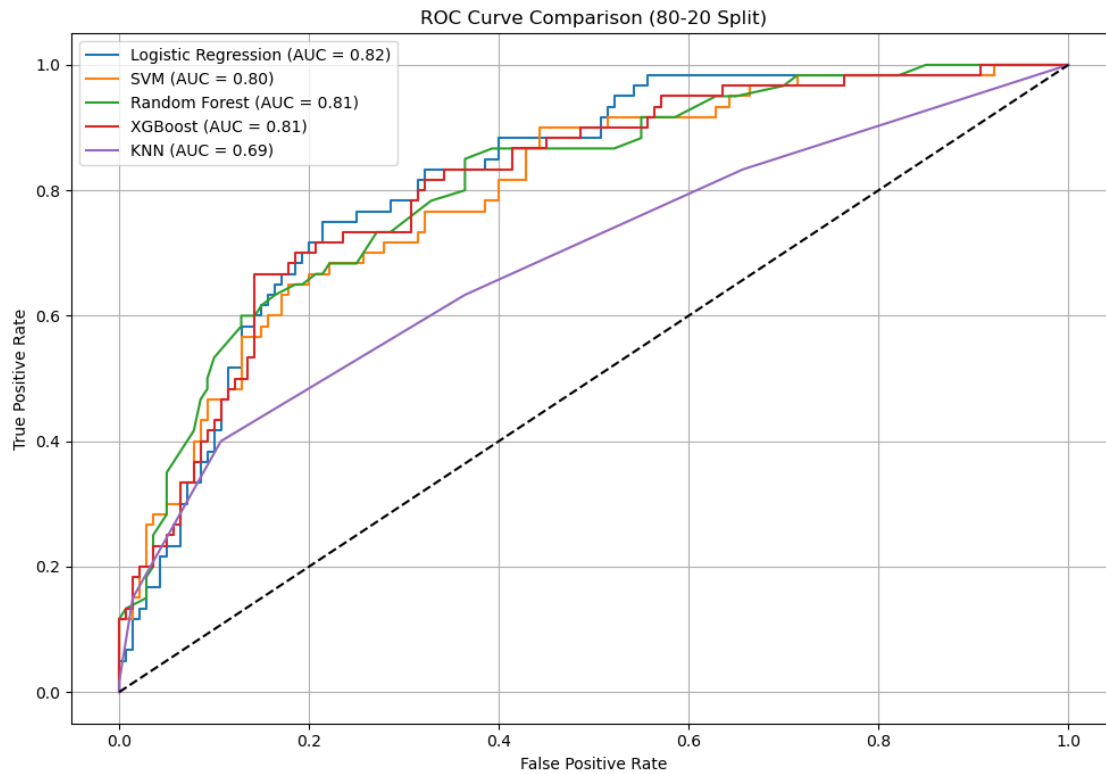
XGBoost :

	precision	recall	f1-score	support
0	0.8121	0.8643	0.8374	140
1	0.6275	0.5333	0.5766	60
accuracy			0.7650	200
macro avg	0.7198	0.6988	0.7070	200
weighted avg	0.7567	0.7650	0.7591	200

KNN :

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

	0	0.7764	0.8929	0.8306	140
	1	0.6154	0.4000	0.4848	60
accuracy				0.7450	200
macro avg		0.6959	0.6464	0.6577	200
weighted avg		0.7281	0.7450	0.7268	200



Summary

```
[13]: print(pd.DataFrame(results))
```

	Logistic Regression	SVM	Random Forest	XGBoost	KNN
Accuracy	0.770000	0.760000	0.770000	0.765000	0.745000
Precision	0.652174	0.772727	0.733333	0.627451	0.615385
Recall	0.500000	0.283333	0.366667	0.533333	0.400000
F1 Score	0.566038	0.414634	0.488889	0.576577	0.484848
AUC	0.818571	0.799405	0.811012	0.810595	0.688631

```
[ ]:
```

Using 5-fold cross validation

```
[14]: cv_results = []

for seed in range(20,40,2):
    cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=seed)

    for name, model in models.items():
        pipeline = Pipeline(steps=[
            ('preprocessor', preprocessor),
            ('classifier', model)
        ])

        auc_scores = cross_val_score(pipeline, X, y, cv=cv, scoring='roc_auc')

        for fold_idx, score in enumerate(auc_scores):
            cv_results.append({
                'model': name,
                'seed': seed,
                'fold': fold_idx,
                'auc': score
            })
```

```
[15]: df_cv_scores = pd.DataFrame(cv_results)
df_model_avg = df_cv_scores.groupby('model')['auc'].mean().reset_index()
df_model_avg = df_model_avg.sort_values(by='auc', ascending=False)
print(df_model_avg)
```

	model	auc
1	Logistic Regression	0.787188
2	Random Forest	0.780689
3	SVM	0.780171
4	XGBoost	0.772850
0	KNN	0.684460

Based on the average AUC scores, Logistic Regression outperformed the other models. Random Forest and SVM followed closely behind with comparable performance. XGBoost showed slightly lower performance, while KNN had the lowest AUC score among all models.

```
[ ]:
```